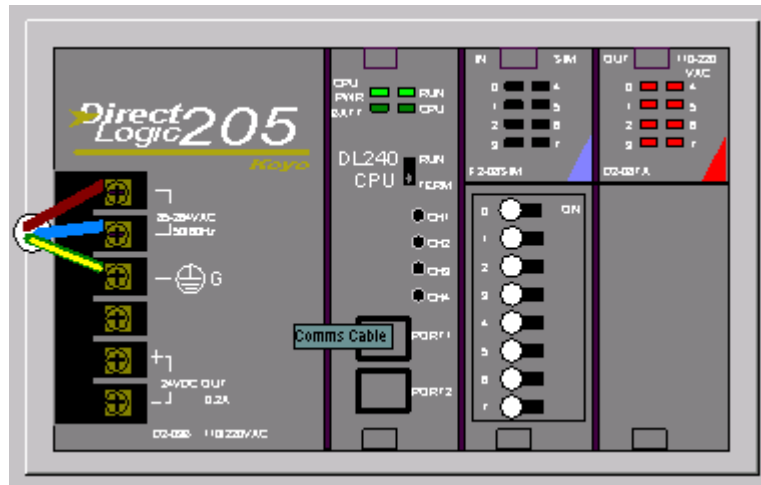


Koyo Direct Logic KDL205KS Protocol driver

Overview

Adroit supports communications with the Koyo Direct Logic DL205 PLC. Via the K-Sequence Protocol. This driver is property Adroit Technologies.



Device configuration

KDL205KS Protocol Driver : BOILER 1

Koyo Direct Logic 205 Details

CPU Model:

Station ID:

Port details

Comm Port:

Baud rate:

Parity:

Stop Bits:

Retries:

Timeout (ms):

OK Cancel Help

DL205 Details

CPU Model: The CPU Model currently installed in the base. 230, 240 or 250.

PLC Station ID: The ID number of the PLC on the PLC network.

Fixed at 1 on Port 1 of the 230 & 240 CPU.

1 – 90 on Port 1 of the 250

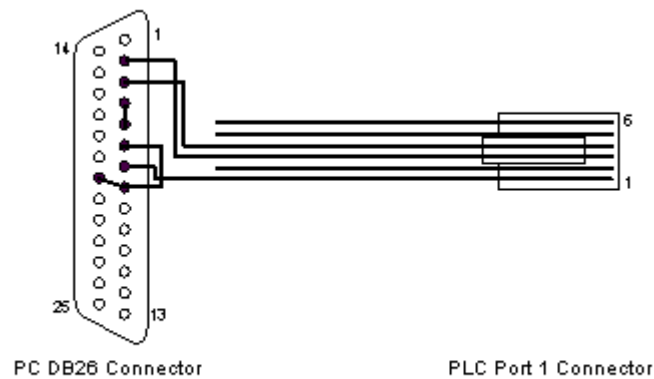
1 – 90 on Port 2 of the 240 & 250.

Port Details

Port:	The available serial ports on your machine
Baud Rate:	The speed of data transmission for the PLC Fixed at 9600 on Port 1 of the 230, 240 & 250. Up to 19200 on Port 2 of the 240 Up to 38400 on Port 2 of the 250
Parity:	Check bit. Fixed at Odd.
Stop Bits:	UART signal bits. Fixed at 1.
Retries:	The number of times that Adroit will try to gain communication with a PLC before failing a transaction.
Timeout:	The time that Adroit will wait for a valid response from a PLC before failing a transaction (Read or Write).

Wiring

The diagram above shows the wiring for the comms cable between a PC serial port and the 230, 240 & 250



CPU port 1 and the 240's Port 2.

For information on the use of port 2 of the 250 please refer to the DL205 User Manual 2nd Edition.

Supported addresses 230 CPU

Prefix	Start	End	Global Start	Global End	Type	Access	Description
TA	0	77	V0	77	WORD	Read Write	Timer Accumulator
CTA	0	77	V1000	1077	WORD	Read Write	Counter Accumulator
			V2000	2377	WORD	Read Write	Variable
			V4000	4177	WORD	Read Write	Variable
			V7260	7647	WORD	Read Write	System Status
			V7750	7777	WORD	Read Write	System Status
X	0	177	V40400	40407	BIT	Read Only	Input
Y	0	177	V40500	40507	BIT	Read Write	Output

C	0	377	V40600	40617	BIT	Read Write	Variable
S	0	377	V41000	41017	BIT	Read Write	Stage Status
T	0	77	V41100	41103	BIT	Read Write	Timer Status
CT	0	77	V41140	41143	BIT	Read Write	Counter Status
SP	0	117	V41200	41204	BIT	Read Only	System Status
SP	540	577	V41226	41227	BIT	Read Only	System Status

Supported addresses 240 CPU

Prefix	Start	End	Global Start	Global End	Type	Access	Description
TA	0	177	V0	77	WORD	Read Write	Timer Accumulator
CTA	0	177	V1000	177	WORD	Read Write	Counter Accumulator
			V2000	3777	WORD	Read Write	Variable
			V4000	4377	WORD	Read Write	Variable
			V7260	7737	WORD	Read Write	System Status
			V7746	7777	WORD	Read Write	System Status
X	0	477	V40400	40423	BIT	Read Only	Input
Y	0	477	V40500	40523	BIT	Read Write	Output
C	0	377	V40600	40617	BIT	Read Write	Variable
S	0	777	V41000	41037	BIT	Read Write	Stage Status
T	0	177	V41100	41107	BIT	Read Write	Timer Status
CT	0	177	V41140	41147	BIT	Read Write	Counter Status
SP	0	137	V41200	41204	BIT	Read Only	System Status
SP	540	617	V41226	41230	BIT	Read Only	System Status

Supported addresses 250 CPU

Prefix	Start	End	Global Start	Global End	Type	Access	Description
TA	0	377	V0	377	WORD	Read Write	Timer Accumulator
CTA	0	177	V1000	1177	WORD	Read Write	Counter Accumulator
			V10000	17777	WORD	Read Write	Variable
			V7400	7777	WORD	Read Write	System Status
X	0	777	V40400	40437	BIT	Read Only	Input
Y	0	777	V40500	40537	BIT	Read Write	Output
C	0	1777	V40600	40677	BIT	Read Write	Variable
S	0	1777	V41000	41077	BIT	Read Write	Stage Status
T	0	377	V41100	41117	BIT	Read Write	Timer Status
CT	0	177	V41140	41147	BIT	Read Write	Counter Status
SP	0	377	V41200	41237	BIT	Read Only	System Status

Adroit Data Types

Supported Data types are as follows:

PLC Data Type	Adroit Data Type		
	Bool	Integer	Real
WORD Read		√	√
WORD Write		√	√
BIT Read	√	√	√
BIT Write	√	√	√

Driver Implementation

NB All addresses with the Koyo DL205 PLC are expressed in Octal.

When reading or writing data to or from the DL205 the driver is actually always using 16 bit words. The driver, depending on the prefix, will work out the type of data and get or put only the correct information. For example when a digital input “X” prefix is used it can be mapped to any Adroit data type. The global equivalent for X0 is V40400. This is a word and X0 to X20 (The next 16 bits) are stored here. This can be read into any Adroit data type except BOOLEAN or any of its derivatives. The most efficient way of reading or writing digitals is to use a multistate agent and read the global equivalent address.

Troubleshooting

During the installation of the driver the following files will be copied:

KDL205KS.DLL	The driver itself.
KDL205KS.PDF	The driver help file.
KDL205KS.WGP	An Adroit database file.
KDL205KS.MIM	An adroit mimic file
KDL205KS.REG	Registry file

Once the driver has been installed and setup, run the REG file, you may then set the Adroit database file to the KDL205KS.WGP and start the agent server.

When the agent server is ready, start the user interface and open the file KDL205KS.MIM. You will then see a representation of a DL205 base unit with a 240 CPU, an input simulation module and an output module.

The database will work with the 230 and 250 as well.

If all is installed correctly and the driver has successfully started, you will be able to see the inputs change when the simulator module switches are changed and also will be able to change output states when the output lights are highlighted and clicked.

This is the first thing to run when first setting up the unit. It can also be used as a training aid to see how the unit works.

Revision Information

Date	Changes
11-Jun-02	Adroit Technologies takes over driver from Alex Griffiths.
14-Aug-03	PDF format document distribution from now on